

Aaran Poon

aaranp1919@gmail.com | +1 236 518 0077 | Vancouver, BC | [linkedin](#) | [Portfolio site](#)
Canadian citizen

EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering, Co-op

May 2026

Vancouver, BC

- Coursework: Computer Architecture, Analog Circuit Design, Power Electronics, and Control System Design; teaching assistant for third-year engineering design studio ELEC 391.

ENGINEERING EXPERIENCE

Synopsys

August 2025 - Present

Verification Engineer | Python, Bash, Bamboo CI, Squish, Linux

Vancouver, BC

- Added 10+ automated tests across Squish GUI and Python command-line workflows for Lumerical FDTD, extending reusable verification utilities across Windows and Linux.
- Execute and analyse approximately 10 test suites per release-validation cycle, reproduce failures, document defects in Jira, and verify fixes through targeted and regression testing.
- Integrate test scripts into Bamboo CI/CD builds, maintain Qt object mappings as the frontend evolves, and use human-in-the-loop AI assistance with manual review of implementations and results.

AMD

September 2024 - December 2024

Graphics Quality Intern | GPU/display validation, C++, Python

Markham, ON

- Performed functional and regression testing of GPU and display hardware/software across configurations, monitoring display output, power draw, temperature fluctuations, and failures while extending coverage for features and bug fixes.
- Investigated black screens and display glitches to distinguish hardware faults from software regressions, debugging C++ source code and developing Python/pywin32 power-cycle automation.
- Built a Python and Streamlit dashboard backed by Microsoft SQL Server to organize and visualize nightly JSON test results by GPU card, test type, and outcome.

CANAL Marine Industries

May 2024 - September 2024

Electrical Engineer Co-op | ETAP, AutoCAD Electrical, CADmatic, Bluebeam

North Vancouver, BC

- Performed shipboard electrical load analysis in ETAP and reviewed cable, transformer, equipment, and power-system requirements to support sizing and design decisions.
- Compared CADmatic 3D models and layouts against revised AutoCAD drawings, updated design information to reflect revisions, and supported vendor/specification reviews and Canadian Coast Guard meetings.

SELECTED DESIGN AND VERIFICATION PROJECTS

ARC4 Decryption and Cracking Hardware | SystemVerilog, Verilog, ModelSim, Quartus, Cyclone V

- Implemented modular RTL for ARC4 state initialization, key scheduling, and pseudo-random generation, plus a parallel brute-force cracker with approximately 2x throughput.
- Developed per-module ModelSim testbenches for RTL and post-synthesis simulation, then validated decrypted output on board against known plaintext.

M68k Computer and Memory Subsystem | Verilog, VHDL, Quartus, DRAM, Cache

- Built coursework-level Verilog and VHDL components including a DRAM controller with refresh and read/write timing, cache subsystem, UART, address decoding, and bus-request/DMA logic.
- Integrated memory and control components within a Motorola 68k computer system using Quartus, applying computer-architecture and FPGA design principles.

Bi-Axial Laser Projector - PCB and FPGA Architecture | Altium, SPICE, FPGA, SPI

- Prototyped and simulated the circuit with SPICE, then designed and laid out a custom two-layer Altium PCB prepared for fabrication.
- Designed synchronous FPGA quadrature-decoder logic with an SPI interface, verified directional tracking in simulation, and selected the simpler STM32 architecture after evaluating the trade-off.

TECHNICAL SKILLS

RTL and Memory Hardware: SystemVerilog, Verilog, VHDL, RTL design, FPGA, DRAM controllers, memory subsystems, cache, address decoding, control logic, ModelSim, Quartus Prime

Circuit Design and Simulation: SPICE simulation, Altium Designer, PCB design and layout, analog circuit design, ETAP, AutoCAD Electrical, CADmatic, Bluebeam

Verification and Automation: functional and regression testing, testbenches, Python, Bash, C, C++, Linux, Git, Bamboo CI, Squish, Jira, pywin32, Streamlit, Microsoft SQL Server